

02_Order_EDA

Purpose: Analyze marketplace order volume, order fulfillment performance, order status distribution, and growth trends to understand business operations and customer purchasing activity.

Q1. How Many Orders Were Placed on the Marketplace?

Result: 99,441

Insight: The marketplace processed 99,441 orders during the analysis period, indicating substantial transaction activity and providing a strong foundation for operational and customer behavior analysis.

Q2. What Is the Distribution of Order Statuses?

Result:

Delivered - 96,478

Shipped - 1,107

Canceled - 625

Unavailable - 609

Invoiced - 314

Processing - 301

Created - 5

Approved - 2

Insight: The marketplace demonstrates strong operational performance, with the vast majority of orders successfully delivered. Only a small percentage of orders were canceled or became unavailable, indicating efficient order fulfillment processes.

Q3. What Percentage of Orders Were Successfully Delivered?

Result: 97.02%

Insight: Approximately 97% of all orders were successfully delivered to customers. This indicates strong fulfillment performance and operational reliability across the marketplace.

Q4. What Percentage of Orders Were Canceled?

Result: 0.63%

Insight: Less than 1% of all orders were canceled, indicating a highly efficient order processing and fulfillment system with minimal transaction loss.

Q5. How Has Order Volume Changed by Year?

Result:

2016 - 329

2017 - 45,101

2018 - 54,011

Insight: Order volume increased significantly from 2017 to 2018, indicating rapid marketplace growth and increasing customer adoption over time. The low order count in 2016 reflects the early stage of marketplace operations.

Q6. How Has Order Volume Changed by Month?

Result:

Order volume increased steadily throughout 2017, reaching a peak of 7,544 orders in November 2017.

Monthly order activity remained consistently strong during 2018, averaging more than 6,000 orders per month until August 2018.

September and October 2018 contain incomplete data and were excluded from trend interpretation.

Insight: The marketplace experienced rapid growth throughout 2017 and maintained strong order volume during 2018, indicating successful customer acquisition and sustained demand.

Q7. Which Month Recorded the Highest Number of Orders?

Result:

November 2017 - 7,544 orders

Insight: November 2017 recorded the highest order volume in the dataset. This spike likely reflects seasonal shopping activity and promotional events that drove increased customer demand.

Q8. Which Day of the Week Receives the Most Orders?

Result:

Monday - 16,196

Tuesday - 15,963

Wednesday - 15,552

Thursday - 14,761

Friday - 14,122

Sunday - 11,960

Saturday - 10,887

Insight: Customer purchasing activity is highest during weekdays, with Monday recording the largest number of orders. Weekend order volume is noticeably lower, indicating that customers are more likely to shop during the workweek.

Q9. What Is the Average Number of Orders Per Day?

Result: 156.85 orders per day

Insight: On average, the marketplace processed approximately 157 orders per day, demonstrating consistent transaction activity and a healthy level of customer engagement throughout the analysis period.

Q10. What Was the Monthly Growth Trend of Orders?

Result:

- Orders increased steadily throughout 2017.
- Monthly orders grew from 800 in January 2017 to 7,544 in November 2017.
- Order volume remained consistently above 6,000 orders per month during most of 2018.
- November 2017 recorded the highest monthly order volume (7,544 orders).
- September and October 2018 contain incomplete data and were excluded from trend interpretation.

Insight: The marketplace experienced strong growth during 2017 and maintained stable transaction volumes throughout 2018, indicating successful customer acquisition and business expansion.